

Homework 5 — Priority-Based Cargo Delivery Simulation

GTU CSE344 System Programming

Dicle Coban

May 2026

Title Page

Course	CSE344 — System Programming
Homework	Homework 5 — Priority-Based Cargo Delivery Simulation
Student	Dicle Coban
Student ID	<i>220104004088</i>
Language	C, POSIX pthreads, C11 atomics
Target Platform	Debian 12, 64-bit POSIX-compatible environment

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1 Introduction

This homework implements a priority-based cargo delivery simulation in C. The program reads delivery orders from an input file, inserts all valid orders into a shared priority queue, creates a fixed number of courier threads, and lets those couriers process orders concurrently until the shift ends.

The main goals of the assignment are:

- to use a fixed-size thread pool instead of creating one thread per order,
- to enforce priority scheduling with `EXPRESS > STANDARD > ECONOMY`,
- to protect the shared queue with mutexes and condition variables,
- to update global statistics with C11 atomic operations,
- to handle `SIGINT` cleanly without interrupting active deliveries,
- to write a structured statistics file after all courier threads terminate.

2 System Design

2.1 High-Level Architecture

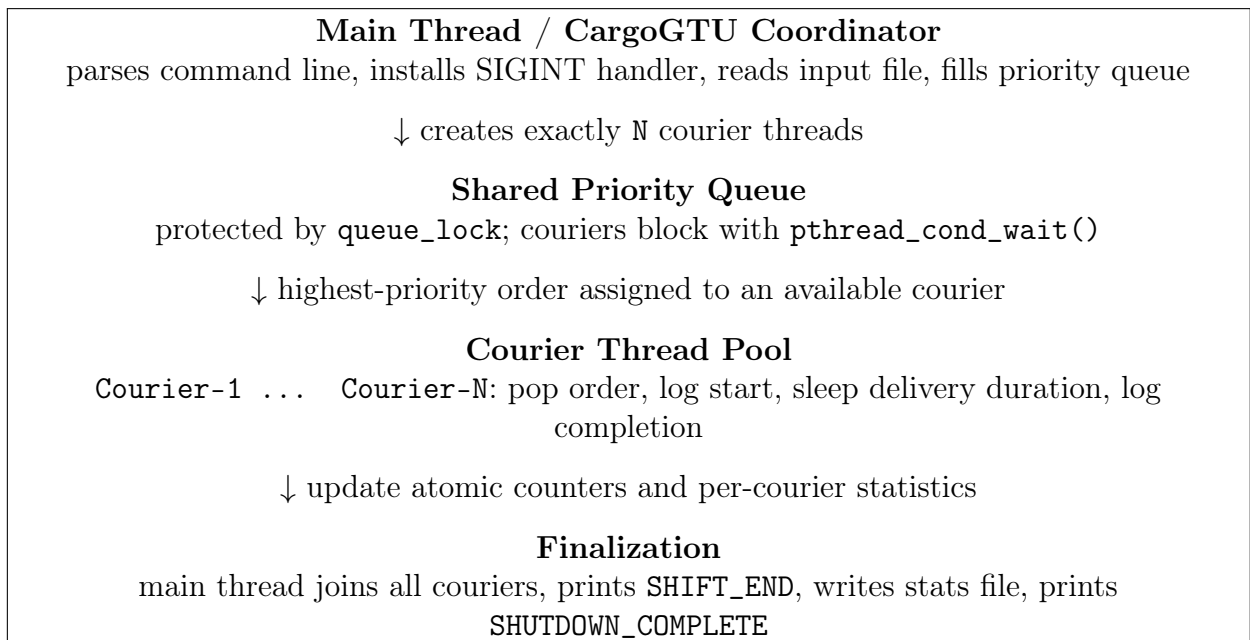


Figure 1: Overall thread-pool and queue architecture.

2.2 Courier Lifecycle

Each courier is created once at startup and is joined once at shutdown. No courier thread is created during order processing. A courier repeatedly performs the following lifecycle:

1. lock `queue_lock`,
2. wait on `courier_wakeup_cond` if the queue is empty but the shift is not over,
3. pop the highest-priority available order,
4. log `DELIVERY_START`,
5. unlock `queue_lock`,
6. simulate delivery by sleeping `duration * 500ms`,
7. update atomic global counters and per-courier statistics,
8. log `DELIVERY_COMPLETE`,
9. notify waiting threads if the shift is complete or more work is available.

When the queue is empty and there are no active deliveries left, the shift enters shutdown mode. Every courier then prints `SHIFT_OVER` and exits normally.

3 Priority Queue Design

The priority queue is implemented as a sorted linked list. The insertion function places each order into the correct position while reading the input file. Therefore, popping from the head always returns the next order that should be delivered.

```
typedef enum {
    PRIORITY_EXPRESS = 1,
    PRIORITY_STANDARD = 2,
    PRIORITY_ECONOMY = 3
} Priority;

typedef struct {
    int id;
    char recipient[33];
    Priority priority;
    int duration_units;
} Order;
```

The comparison rule is:

1. lower numeric priority value comes first, so **EXPRESS** is first,
2. if two orders have the same priority, the order with the lower `id` comes first.

This design keeps `queue_pop()` simple and deterministic. Since all valid input orders are inserted before the courier threads start, every courier sees the same globally ordered work queue.

4 Synchronization Strategy

4.1 Synchronization Primitives

Primitive	Location	Purpose
queue_lock	Main queue and shift state	Protects the priority queue, shift_stopping , and active_delivery_count .
courier_wakeup_cond	Courier waiting path	Blocks idle couriers without busy waiting and wakes them when work or shutdown is available.
shift_finished_cond	Main thread wait path	Lets the main thread wait until the shift is completed or all active deliveries finish after SIGINT.
stdout_lock	Console output	Ensures each stdout log line is printed as one clean line and prevents interleaved output.
_Atomic int total_completed_orders	Global statistics	Counts completed deliveries without using a mutex.
_Atomic int total_cancelled_orders	Global statistics	Counts cancelled pending orders after SIGINT.
_Atomic long total_completed_delivery_time	Global statistics	Accumulates completed delivery durations in milliseconds.

4.2 Avoiding Busy Waiting

Courier threads do not spin when the queue is empty. They use `pthread_cond_wait()` while holding `queue_lock`. The condition is checked in a `while` loop so that spurious wakeups do not cause incorrect behavior.

```
while (queue_size(&pending_order_queue) == 0 &&
      !shift_stopping &&
      active_delivery_count > 0) {
    print_log_line("[COURIER-%d] WAITING\n",
                  courier_arg->courier_id);
    pthread_cond_wait(&courier_wakeup_cond, &queue_lock);
}
```

5 SIGINT Handling

The program uses a minimal signal handler. The handler only sets a `volatile sig_atomic_t` flag, which is safe for signal context. The actual shutdown logic is performed later in normal program flow by the main thread.

```
static void sigint_handler(int signo)
{
```

```
(void)signo;  
sigint_flag = 1;  
}
```

The shutdown sequence after `SIGINT` is:

1. the signal handler sets `sigint_flag`,
2. the main thread notices the flag while waiting for shift completion,
3. `shift_stopping` is set under `queue_lock`,
4. pending orders still in the queue are removed and logged as `ORDER_CANCELLED`,
5. `total_cancelled_orders` is updated with atomic operations,
6. waiting couriers are awakened with `pthread_cond_broadcast()`,
7. active couriers finish their current deliveries without interruption,
8. all courier threads are joined,
9. `SHIFT_END`, the statistics file, and `SHUTDOWN_COMPLETE` are produced.

This design prevents unsafe work inside the signal handler and preserves the rule that an active delivery cannot be interrupted midway.

6 Input Parsing and Output Format

The input parser reads each line in the format:

```
<id> <name> <priority> <duration>
```

Blank lines and malformed lines are skipped silently. Valid orders require a positive `id`, a positive duration, an ASCII recipient name made of letters, digits, and underscores, and one of the accepted priority strings: `EXPRESS`, `STANDARD`, or `ECONOMY`.

Console output is protected by `stdout_lock`. The statistics file is written once after all courier threads have been joined. It contains total order count, completed count, cancelled count, total simulated delivery time, average time per completed order, and per-courier statistics.

7 Test Scenarios

The following scenarios were run according to the homework specification. Each screenshot in the final report is uncut and shows the command prompt, command, program output, and program termination.

7.1 Scenario 1 — Mixed Orders with 3 Couriers

```
./cargoGTU -n 3 -i tests/10orders_mix.txt -s stats.txt
cat stats.txt
```

Result: all 10 orders completed, no order was cancelled, and the statistics file shows `total_time=58000ms`.

```
diclesaracoban@Dicle-MacBook-Pro HW5 % ./cargoGTU -n 3 -i tests/10orders_mix.txt -s stats.txt
cat stats.txt
[CARGOGTU] SHIFT_START couriers=3 orders=10
[CARGOGTU] ORDER_QUEUED id=1 recipient=Ahmet_Yilmaz priority=EXPRESS duration=8
[CARGOGTU] ORDER_QUEUED id=4 recipient=Fatma_Sahin priority=EXPRESS duration=6
[CARGOGTU] ORDER_QUEUED id=7 recipient=Mustafa_Koc priority=EXPRESS duration=7
[CARGOGTU] ORDER_QUEUED id=2 recipient=Ayşe_Kaya priority=STANDARD duration=12
[CARGOGTU] ORDER_QUEUED id=5 recipient=Ali_Celik priority=STANDARD duration=9
[CARGOGTU] ORDER_QUEUED id=8 recipient=Elif_Demir priority=STANDARD duration=11
[CARGOGTU] ORDER_QUEUED id=10 recipient=Merve_Yildiz priority=STANDARD duration=10
[CARGOGTU] ORDER_QUEUED id=3 recipient=Mehmet_Demir priority=ECONOMY duration=20
[CARGOGTU] ORDER_QUEUED id=6 recipient=Zeynep_Arslan priority=ECONOMY duration=15
[CARGOGTU] ORDER_QUEUED id=9 recipient=Hasan_Ozturk priority=ECONOMY duration=18
[COURIER-1] DELIVERY_START id=1 recipient=Ahmet_Yilmaz priority=EXPRESS
[COURIER-2] DELIVERY_START id=4 recipient=Fatma_Sahin priority=EXPRESS
[COURIER-3] DELIVERY_START id=7 recipient=Mustafa_Koc priority=EXPRESS
[COURIER-2] DELIVERY_COMPLETE id=4 recipient=Fatma_Sahin duration=3000ms
[COURIER-2] DELIVERY_START id=2 recipient=Ayşe_Kaya priority=STANDARD
[COURIER-3] DELIVERY_COMPLETE id=7 recipient=Mustafa_Koc duration=3500ms
[COURIER-3] DELIVERY_START id=5 recipient=Ali_Celik priority=STANDARD
[COURIER-1] DELIVERY_COMPLETE id=1 recipient=Ahmet_Yilmaz duration=4000ms
[COURIER-1] DELIVERY_START id=8 recipient=Elif_Demir priority=STANDARD
[COURIER-3] DELIVERY_COMPLETE id=5 recipient=Ali_Celik duration=4500ms
[COURIER-3] DELIVERY_START id=10 recipient=Merve_Yildiz priority=STANDARD
[COURIER-2] DELIVERY_COMPLETE id=2 recipient=Ayşe_Kaya duration=6000ms
[COURIER-2] DELIVERY_START id=3 recipient=Mehmet_Demir priority=ECONOMY
[COURIER-1] DELIVERY_COMPLETE id=8 recipient=Elif_Demir duration=5500ms
[COURIER-1] DELIVERY_START id=6 recipient=Zeynep_Arslan priority=ECONOMY
[COURIER-3] DELIVERY_COMPLETE id=10 recipient=Merve_Yildiz duration=5000ms
[COURIER-3] DELIVERY_START id=9 recipient=Hasan_Ozturk priority=ECONOMY
[COURIER-1] DELIVERY_COMPLETE id=6 recipient=Zeynep_Arslan duration=7500ms
[COURIER-1] WAITING
[COURIER-2] DELIVERY_COMPLETE id=3 recipient=Mehmet_Demir duration=10000ms
[COURIER-2] WAITING
[COURIER-3] DELIVERY_COMPLETE id=9 recipient=Hasan_Ozturk duration=9000ms
[COURIER-3] SHIFT_OVER
[COURIER-1] SHIFT_OVER
[COURIER-2] SHIFT_OVER
[CARGOGTU] SHIFT_END completed=10 cancelled=0 total_time=58000ms
[CARGOGTU] SHUTDOWN_COMPLETE
SHIFT_SUMMARY
Total orders      : 10
Completed         : 10
Cancelled         : 0
Total time        : 58000ms
Avg per order     : 5800ms

COURIER_STATS
Courier-1  completed=3  total_time=17000ms
Courier-2  completed=3  total_time=19000ms
Courier-3  completed=4  total_time=22000ms
```

Figure 2: Scenario 1 console output: `10orders_mix.txt` with 3 couriers and final `stats.txt`.

7.2 Scenario 2 — Economy Orders with 10 Couriers

```
./cargoGTU -n 10 -i tests/10orders_economy.txt -s stats.txt
cat stats.txt
```

Result: all 10 orders completed exactly once. Since all orders have the same priority, the DELIVERY_START order follows increasing order id.

```
diclesaracoban@Dicle-MacBook-Pro HW5 % ./cargoGTU -n 10 -i tests/10orders_economy.txt -s stats.txt
cat stats.txt
[CARGOGTU] SHIFT_START couriers=10 orders=10
[CARGOGTU] ORDER_QUEUED id=1 recipient=Ahmet_Yilmaz priority=ECONOMY duration=8
[CARGOGTU] ORDER_QUEUED id=2 recipient=Ayse_Kaya priority=ECONOMY duration=10
[CARGOGTU] ORDER_QUEUED id=3 recipient=Mehmet_Demir priority=ECONOMY duration=7
[CARGOGTU] ORDER_QUEUED id=4 recipient=Fatma_Sahin priority=ECONOMY duration=12
[CARGOGTU] ORDER_QUEUED id=5 recipient=Ali_Celik priority=ECONOMY duration=9
[CARGOGTU] ORDER_QUEUED id=6 recipient=Zeynep_Arslan priority=ECONOMY duration=11
[CARGOGTU] ORDER_QUEUED id=7 recipient=Mustafa_Koc priority=ECONOMY duration=8
[CARGOGTU] ORDER_QUEUED id=8 recipient=Elif_Demir priority=ECONOMY duration=6
[CARGOGTU] ORDER_QUEUED id=9 recipient=Hasan_Ozturk priority=ECONOMY duration=10
[CARGOGTU] ORDER_QUEUED id=10 recipient=Merve_Yildiz priority=ECONOMY duration=9
[COURIER-1] DELIVERY_START id=1 recipient=Ahmet_Yilmaz priority=ECONOMY
[COURIER-3] DELIVERY_START id=2 recipient=Ayse_Kaya priority=ECONOMY
[COURIER-4] DELIVERY_START id=3 recipient=Mehmet_Demir priority=ECONOMY
[COURIER-5] DELIVERY_START id=4 recipient=Fatma_Sahin priority=ECONOMY
[COURIER-2] DELIVERY_START id=5 recipient=Ali_Celik priority=ECONOMY
[COURIER-7] DELIVERY_START id=6 recipient=Zeynep_Arslan priority=ECONOMY
[COURIER-8] DELIVERY_START id=7 recipient=Mustafa_Koc priority=ECONOMY
[COURIER-9] DELIVERY_START id=8 recipient=Elif_Demir priority=ECONOMY
[COURIER-10] DELIVERY_START id=9 recipient=Hasan_Ozturk priority=ECONOMY
[COURIER-6] DELIVERY_START id=10 recipient=Merve_Yildiz priority=ECONOMY
[COURIER-9] DELIVERY_COMPLETE id=8 recipient=Elif_Demir duration=3000ms
[COURIER-9] WAITING
[COURIER-4] DELIVERY_COMPLETE id=3 recipient=Mehmet_Demir duration=3500ms
[COURIER-4] WAITING
[COURIER-1] DELIVERY_COMPLETE id=1 recipient=Ahmet_Yilmaz duration=4000ms
[COURIER-1] WAITING
[COURIER-8] DELIVERY_COMPLETE id=7 recipient=Mustafa_Koc duration=4000ms
[COURIER-8] WAITING
[COURIER-2] DELIVERY_COMPLETE id=5 recipient=Ali_Celik duration=4500ms
[COURIER-2] WAITING
[COURIER-6] DELIVERY_COMPLETE id=10 recipient=Merve_Yildiz duration=4500ms
[COURIER-6] WAITING
[COURIER-3] DELIVERY_COMPLETE id=2 recipient=Ayse_Kaya duration=5000ms
[COURIER-3] WAITING
[COURIER-10] DELIVERY_COMPLETE id=9 recipient=Hasan_Ozturk duration=5000ms
[COURIER-10] WAITING
[COURIER-7] DELIVERY_COMPLETE id=6 recipient=Zeynep_Arslan duration=5500ms
[COURIER-7] WAITING
[COURIER-5] DELIVERY_COMPLETE id=4 recipient=Fatma_Sahin duration=6000ms
[COURIER-5] SHIFT_OVER
[COURIER-9] SHIFT_OVER
[COURIER-4] SHIFT_OVER
[COURIER-1] SHIFT_OVER
[COURIER-8] SHIFT_OVER
[COURIER-2] SHIFT_OVER
[COURIER-6] SHIFT_OVER
[COURIER-3] SHIFT_OVER
[COURIER-10] SHIFT_OVER
[COURIER-7] SHIFT_OVER
[CARGOGTU] SHIFT_END completed=10 cancelled=0 total_time=45000ms
[CARGOGTU] SHUTDOWN_COMPLETE
SHIFT_SUMMARY
Total orders      : 10
Completed         : 10
Cancelled         : 0
Total time        : 45000ms
Avg per order     : 4500ms

COURIER_STATS
Courier-1 completed=1 total_time=4000ms
Courier-2 completed=1 total_time=4500ms
Courier-3 completed=1 total_time=5000ms
Courier-4 completed=1 total_time=3500ms
Courier-5 completed=1 total_time=6000ms
Courier-6 completed=1 total_time=4500ms
Courier-7 completed=1 total_time=5500ms
Courier-8 completed=1 total_time=4000ms
Courier-9 completed=1 total_time=3000ms
Courier-10 completed=1 total_time=5000ms
```

Figure 3: Scenario 2 console output: 10orders_economy.txt with 10 couriers and final summary.

7.3 Scenario 3 — SIGINT Shutdown

```
./cargoGTU -n 2 -i tests/20orders_mix.txt -s stats.txt
```

After approximately two seconds, **Ctrl+C** was sent. Active deliveries completed, pending orders were cancelled, and the final summary showed that completed plus cancelled equaled the total number of valid orders.

```
cat stats.txt  
ps aux | grep cargoGTU
```

```
● diclesaracoban@Dicle-MacBook-Pro HW5 % ./cargoGTU -n 2 -i tests/20orders_mix.txt -s stats.txt  
[CARGOGTU] SHIFT_START couriers=2 orders=20  
[CARGOGTU] ORDER_QUEUED id=1 recipient=Ahmet_Yilmaz priority=EXPRESS duration=1  
[CARGOGTU] ORDER_QUEUED id=4 recipient=Fatma_Sahin priority=EXPRESS duration=1  
[CARGOGTU] ORDER_QUEUED id=7 recipient=Mustafa_Koc priority=EXPRESS duration=1  
[CARGOGTU] ORDER_QUEUED id=10 recipient=Merve_Yildiz priority=EXPRESS duration=1  
[CARGOGTU] ORDER_QUEUED id=13 recipient=Burak_Arslan priority=EXPRESS duration=1  
[CARGOGTU] ORDER_QUEUED id=16 recipient=Rana_Yilmaz priority=EXPRESS duration=2  
[CARGOGTU] ORDER_QUEUED id=19 recipient=Oguz_Yildiz priority=EXPRESS duration=3  
[CARGOGTU] ORDER_QUEUED id=3 recipient=Mehmet_Demir priority=STANDARD duration=2  
[CARGOGTU] ORDER_QUEUED id=5 recipient=Ali_Celik priority=STANDARD duration=3  
[CARGOGTU] ORDER_QUEUED id=8 recipient=Elif_Demir priority=STANDARD duration=2  
[CARGOGTU] ORDER_QUEUED id=11 recipient=Emre_Celik priority=STANDARD duration=2  
[CARGOGTU] ORDER_QUEUED id=14 recipient=Dilan_Ozkan priority=STANDARD duration=2  
[CARGOGTU] ORDER_QUEUED id=17 recipient=Tarik_Demir priority=STANDARD duration=1  
[CARGOGTU] ORDER_QUEUED id=20 recipient=Ceren_Arslan priority=STANDARD duration=1  
[CARGOGTU] ORDER_QUEUED id=2 recipient=Ayşe_Kaya priority=ECONOMY duration=3  
[CARGOGTU] ORDER_QUEUED id=6 recipient=Zeynep_Arslan priority=ECONOMY duration=2  
[CARGOGTU] ORDER_QUEUED id=9 recipient=Hasan_Ozturk priority=ECONOMY duration=3  
[CARGOGTU] ORDER_QUEUED id=12 recipient=Selin_Kaya priority=ECONOMY duration=3  
[CARGOGTU] ORDER_QUEUED id=15 recipient=Kemal_Sahin priority=ECONOMY duration=3  
[CARGOGTU] ORDER_QUEUED id=18 recipient=Pinar_Koc priority=ECONOMY duration=2  
[COURIER-1] DELIVERY_START id=1 recipient=Ahmet_Yilmaz priority=EXPRESS  
[COURIER-2] DELIVERY_START id=4 recipient=Fatma_Sahin priority=EXPRESS  
^C[CARGOGTU] SIGINT_RECEIVED pending_orders=18  
[CARGOGTU] ORDER_CANCELLED id=7 recipient=Mustafa_Koc priority=EXPRESS  
[CARGOGTU] ORDER_CANCELLED id=10 recipient=Merve_Yildiz priority=EXPRESS  
[CARGOGTU] ORDER_CANCELLED id=13 recipient=Burak_Arslan priority=EXPRESS  
[CARGOGTU] ORDER_CANCELLED id=16 recipient=Rana_Yilmaz priority=EXPRESS  
[CARGOGTU] ORDER_CANCELLED id=19 recipient=Oguz_Yildiz priority=EXPRESS  
[CARGOGTU] ORDER_CANCELLED id=3 recipient=Mehmet_Demir priority=STANDARD  
[CARGOGTU] ORDER_CANCELLED id=5 recipient=Ali_Celik priority=STANDARD  
[CARGOGTU] ORDER_CANCELLED id=8 recipient=Elif_Demir priority=STANDARD  
[CARGOGTU] ORDER_CANCELLED id=11 recipient=Emre_Celik priority=STANDARD  
[CARGOGTU] ORDER_CANCELLED id=14 recipient=Dilan_Ozkan priority=STANDARD  
[CARGOGTU] ORDER_CANCELLED id=17 recipient=Tarik_Demir priority=STANDARD  
[CARGOGTU] ORDER_CANCELLED id=20 recipient=Ceren_Arslan priority=STANDARD  
[CARGOGTU] ORDER_CANCELLED id=2 recipient=Ayşe_Kaya priority=ECONOMY  
[CARGOGTU] ORDER_CANCELLED id=6 recipient=Zeynep_Arslan priority=ECONOMY  
[CARGOGTU] ORDER_CANCELLED id=9 recipient=Hasan_Ozturk priority=ECONOMY  
[CARGOGTU] ORDER_CANCELLED id=12 recipient=Selin_Kaya priority=ECONOMY  
[CARGOGTU] ORDER_CANCELLED id=15 recipient=Kemal_Sahin priority=ECONOMY  
[CARGOGTU] ORDER_CANCELLED id=18 recipient=Pinar_Koc priority=ECONOMY  
[COURIER-1] DELIVERY_COMPLETE id=1 recipient=Ahmet_Yilmaz duration=500ms  
[COURIER-1] SHIFT_OVER  
[COURIER-2] DELIVERY_COMPLETE id=4 recipient=Fatma_Sahin duration=500ms  
[COURIER-2] SHIFT_OVER  
[CARGOGTU] SHIFT_END completed=2 cancelled=18 total_time=1000ms  
[CARGOGTU] SHUTDOWN_COMPLETE
```

Figure 4: Scenario 3 SIGINT output: **Ctrl+C** triggers cancellation of pending orders while active deliveries complete.

```

[diclesaracoba] ~/cargoGTU % cat stats.txt
diclesaracoba@Dicle-MacBook-Pro HW5 % cat stats.txt
ps aux | grep cargoGTU
SHIFT_SUMMARY
Total orders      : 20
Completed         : 2
Cancelled         : 18
Total time        : 1000ms
Avg per order     : 500ms

COURIER_STATS
Courier-1 completed=1 total_time=500ms
Courier-2 completed=1 total_time=500ms
diclesaracoba  68426  0.0  0.0 416631504  176 s001  U+   1:12AM  0:00.00 grep cargoGTU

```

Figure 5: Scenario 3 statistics and process check: Final `stats.txt` and `ps` check after SIGINT shutdown.

7.4 Scenario 4 — Valgrind

```

docker run --rm -v "$PWD":/work -w /work debian:12 bash -lc \
'apt-get update >/dev/null &&
apt-get install -y build-essential valgrind make >/dev/null &&
make clean && make &&
valgrind --leak-check=full --show-leak-kinds=all \
./cargoGTU -n 3 -i tests/10orders_mix.txt -s stats_valgrind.txt'

```

Result: the program was compiled and executed inside a Debian 12 Docker container with Valgrind. The run reported 0 bytes in 0 blocks at exit, All heap blocks were freed, and ERROR SUMMARY: 0 errors.

```

diclesaracoban@Dicle-MacBook-Pro HW5 % docker run --rm -v "$PWD":/work -w /work debian:12 bash -lc 'apt-get
update >/dev/null && apt-get install -y build-essential valgrind make >/dev/null && make clean && make &&
valgrind --leak-check=full --show-leak-kinds=all ./cargoGTU -n 3 -i tests/10orders_mix.txt -s stats_valgrin
d.txt'
debconf: delaying package configuration, since apt-utils is not installed
rm -f cargoGTU main.o order_queue.o
gcc -Wall -Wextra -std=c11 -pthread -c main.c
gcc -Wall -Wextra -std=c11 -pthread -c order_queue.c
gcc -Wall -Wextra -std=c11 -pthread -o cargoGTU main.o order_queue.o
==1== Memcheck, a memory error detector
==1== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==1== Using Valgrind-3.19.0 and LibVEX; rerun with -h for copyright info
==1== Command: ./cargoGTU -n 3 -i tests/10orders_mix.txt -s stats_valgrind.txt
==1==
[CARGOGTU] SHIFT_START couriers=3 orders=10
[CARGOGTU] ORDER_QUEUE id=1 recipient=Ahmet_Yilmaz priority=EXPRESS duration=8
[CARGOGTU] ORDER_QUEUE id=4 recipient=Fatma_Sahin priority=EXPRESS duration=6
[CARGOGTU] ORDER_QUEUE id=7 recipient=Mustafa_Koc priority=EXPRESS duration=7
[CARGOGTU] ORDER_QUEUE id=2 recipient=Ayşe_Kaya priority=STANDARD duration=12
[CARGOGTU] ORDER_QUEUE id=5 recipient=Ali_Celik priority=STANDARD duration=9
[CARGOGTU] ORDER_QUEUE id=8 recipient=Elif_Demir priority=STANDARD duration=11
[CARGOGTU] ORDER_QUEUE id=10 recipient=Merve_Yildiz priority=STANDARD duration=10
[CARGOGTU] ORDER_QUEUE id=3 recipient=Mehmet_Demir priority=ECONOMY duration=20
[CARGOGTU] ORDER_QUEUE id=6 recipient=Zeynep_Arslan priority=ECONOMY duration=15
[CARGOGTU] ORDER_QUEUE id=9 recipient=Hasan_Ozturk priority=ECONOMY duration=18
[COURIER-1] DELIVERY_START id=1 recipient=Ahmet_Yilmaz priority=EXPRESS
[COURIER-2] DELIVERY_START id=4 recipient=Fatma_Sahin priority=EXPRESS
[COURIER-3] DELIVERY_START id=7 recipient=Mustafa_Koc priority=EXPRESS
[COURIER-2] DELIVERY_COMPLETE id=4 recipient=Fatma_Sahin duration=3000ms
[COURIER-2] DELIVERY_START id=2 recipient=Ayşe_Kaya priority=STANDARD
[COURIER-3] DELIVERY_COMPLETE id=7 recipient=Mustafa_Koc duration=3500ms
[COURIER-3] DELIVERY_START id=5 recipient=Ali_Celik priority=STANDARD
[COURIER-1] DELIVERY_COMPLETE id=1 recipient=Ahmet_Yilmaz duration=4000ms
[COURIER-1] DELIVERY_START id=8 recipient=Elif_Demir priority=STANDARD
[COURIER-3] DELIVERY_COMPLETE id=5 recipient=Ali_Celik duration=4500ms
[COURIER-3] DELIVERY_START id=10 recipient=Merve_Yildiz priority=STANDARD
[COURIER-2] DELIVERY_COMPLETE id=2 recipient=Ayşe_Kaya duration=6000ms
[COURIER-2] DELIVERY_START id=3 recipient=Mehmet_Demir priority=ECONOMY
[COURIER-1] DELIVERY_COMPLETE id=8 recipient=Elif_Demir duration=5500ms
[COURIER-1] DELIVERY_START id=6 recipient=Zeynep_Arslan priority=ECONOMY
[COURIER-3] DELIVERY_COMPLETE id=10 recipient=Merve_Yildiz duration=5000ms
[COURIER-3] DELIVERY_START id=9 recipient=Hasan_Ozturk priority=ECONOMY
[COURIER-1] DELIVERY_COMPLETE id=6 recipient=Zeynep_Arslan duration=7500ms
[COURIER-1] WAITING
[COURIER-2] DELIVERY_COMPLETE id=3 recipient=Mehmet_Demir duration=10000ms
[COURIER-2] WAITING
[COURIER-3] DELIVERY_COMPLETE id=9 recipient=Hasan_Ozturk duration=9000ms
[COURIER-3] SHIFT_OVER
[COURIER-1] SHIFT_OVER
[COURIER-2] SHIFT_OVER
[CARGOGTU] SHIFT_END completed=10 cancelled=0 total_time=58000ms
[CARGOGTU] SHUTDOWN_COMPLETE
==1==
==1== HEAP SUMMARY:
==1==   in use at exit: 0 bytes in 0 blocks
==1==   total heap usage: 21 allocs, 21 frees, 14,692 bytes allocated
==1==
==1== All heap blocks were freed -- no leaks are possible
==1==
==1== For lists of detected and suppressed errors, rerun with: -s
==1== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)

```

Figure 6: Scenario 4 Valgrind output: Debian 12 Docker run showing 0 errors and no leaks.

7.5 Scenario 5 — Multiple High-Concurrency Runs

```

./cargoGTU -n 10 -i tests/20orders_mix.txt -s stats.txt | tee test5_run1.txt
./cargoGTU -n 10 -i tests/20orders_mix.txt -s stats.txt | tee test5_run2.txt
./cargoGTU -n 10 -i tests/20orders_mix.txt -s stats.txt | tee test5_run3.txt
grep "SHIFT_END" test5_run*.txt
grep "DELIVERY_START" test5_run1.txt | awk '{for(i=1;i<=NF;i++) if($i ~ /^id=/)
    print $i}' | sort | uniq -d
cat stats.txt

```

Result: every captured run ends with `completed=20`, `cancelled=0`, and `total_time=19500ms`. The screenshots are consecutive parts of the long high-concurrency console output. The final `duplicate-id` command is expected to produce no output; this means that no order id appears more than once among the `DELIVERY_START` lines.

```
diclesaracoban@Dicle-MacBook-Pro HW5 % ./cargoGTU -n 10 -i tests/20orders_mix.txt -s stats.txt | tee test5_run1.txt
./cargoGTU -n 10 -i tests/20orders_mix.txt -s stats.txt | tee test5_run2.txt
./cargoGTU -n 10 -i tests/20orders_mix.txt -s stats.txt | tee test5_run3.txt
grep "SHIFT_END" test5_run*.txt
grep "DELIVERY_START" test5_run1.txt | awk '{for(i=1;i<=NF;i++) if($i ~ /^id=/) print $i}' | sort | uniq -d
cat stats.txt
[CARGOGTU] SHIFT_START couriers=10 orders=20
[CARGOGTU] ORDER_QUEUED id=1 recipient=Ahmet_Yilmaz priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=4 recipient=Fatma_Sahin priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=7 recipient=Mustafa_Koc priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=10 recipient=Merve_Yildiz priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=13 recipient=Burak_Arslan priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=16 recipient=Rana_Yilmaz priority=EXPRESS duration=2
[CARGOGTU] ORDER_QUEUED id=19 recipient=Oguz_Yildiz priority=EXPRESS duration=3
[CARGOGTU] ORDER_QUEUED id=3 recipient=Mehmet_Demir priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=5 recipient=Ali_Celik priority=STANDARD duration=3
[CARGOGTU] ORDER_QUEUED id=8 recipient=Elif_Demir priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=11 recipient=Emre_Celik priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=14 recipient=Dilan_Ozkan priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=17 recipient=Tarik_Demir priority=STANDARD duration=1
[CARGOGTU] ORDER_QUEUED id=20 recipient=Ceren_Arslan priority=STANDARD duration=1
[CARGOGTU] ORDER_QUEUED id=2 recipient=Ayşe_Kaya priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=6 recipient=Zeynep_Arslan priority=ECONOMY duration=2
[CARGOGTU] ORDER_QUEUED id=9 recipient=Hasan_Ozturk priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=12 recipient=Selin_Kaya priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=15 recipient=Kemal_Sahin priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=18 recipient=Pinar_Koc priority=ECONOMY duration=2
[COURIER-1] DELIVERY_START id=1 recipient=Ahmet_Yilmaz priority=EXPRESS
[COURIER-2] DELIVERY_START id=4 recipient=Fatma_Sahin priority=EXPRESS
[COURIER-6] DELIVERY_START id=7 recipient=Mustafa_Koc priority=EXPRESS
[COURIER-3] DELIVERY_START id=10 recipient=Merve_Yildiz priority=EXPRESS
[COURIER-9] DELIVERY_START id=13 recipient=Burak_Arslan priority=EXPRESS
[COURIER-4] DELIVERY_START id=16 recipient=Rana_Yilmaz priority=EXPRESS
[COURIER-10] DELIVERY_START id=19 recipient=Oguz_Yildiz priority=EXPRESS
[COURIER-8] DELIVERY_START id=3 recipient=Mehmet_Demir priority=STANDARD
[COURIER-7] DELIVERY_START id=5 recipient=Ali_Celik priority=STANDARD
[COURIER-5] DELIVERY_START id=8 recipient=Elif_Demir priority=STANDARD
[COURIER-6] DELIVERY_COMPLETE id=7 recipient=Mustafa_Koc duration=500ms
[COURIER-1] DELIVERY_COMPLETE id=1 recipient=Ahmet_Yilmaz duration=500ms
[COURIER-6] DELIVERY_START id=11 recipient=Emre_Celik priority=STANDARD
[COURIER-1] DELIVERY_START id=14 recipient=Dilan_Ozkan priority=STANDARD
[COURIER-3] DELIVERY_COMPLETE id=10 recipient=Merve_Yildiz duration=500ms
[COURIER-3] DELIVERY_START id=17 recipient=Tarik_Demir priority=STANDARD
[COURIER-2] DELIVERY_COMPLETE id=4 recipient=Fatma_Sahin duration=500ms
[COURIER-2] DELIVERY_START id=20 recipient=Ceren_Arslan priority=STANDARD
[COURIER-9] DELIVERY_COMPLETE id=13 recipient=Burak_Arslan duration=500ms
[COURIER-9] DELIVERY_START id=2 recipient=Ayşe_Kaya priority=ECONOMY
[COURIER-4] DELIVERY_COMPLETE id=16 recipient=Rana_Yilmaz duration=1000ms
[COURIER-4] DELIVERY_START id=6 recipient=Zeynep_Arslan priority=ECONOMY
[COURIER-8] DELIVERY_COMPLETE id=3 recipient=Mehmet_Demir duration=1000ms
[COURIER-8] DELIVERY_START id=9 recipient=Hasan_Ozturk priority=ECONOMY
[COURIER-5] DELIVERY_COMPLETE id=17 recipient=Tarik_Demir duration=500ms
[COURIER-3] DELIVERY_START id=12 recipient=Selin_Kaya priority=ECONOMY
[COURIER-5] DELIVERY_COMPLETE id=8 recipient=Elif_Demir duration=1000ms
[COURIER-5] DELIVERY_START id=15 recipient=Kemal_Sahin priority=ECONOMY
[COURIER-2] DELIVERY_COMPLETE id=20 recipient=Ceren_Arslan duration=500ms
[COURIER-2] DELIVERY_START id=18 recipient=Pinar_Koc priority=ECONOMY
[COURIER-10] DELIVERY_COMPLETE id=19 recipient=Oguz_Yildiz duration=1500ms
[COURIER-10] WAITING
[COURIER-7] DELIVERY_COMPLETE id=5 recipient=Ali_Celik duration=1500ms
[COURIER-7] WAITING
[COURIER-6] DELIVERY_COMPLETE id=11 recipient=Emre_Celik duration=1000ms
[COURIER-6] WAITING
[COURIER-1] DELIVERY_COMPLETE id=14 recipient=Dilan_Ozkan duration=1000ms
[COURIER-1] WAITING
[COURIER-4] DELIVERY_COMPLETE id=6 recipient=Zeynep_Arslan duration=1000ms
[COURIER-4] WAITING
[COURIER-9] DELIVERY_COMPLETE id=2 recipient=Ayşe_Kaya duration=1500ms
[COURIER-9] WAITING
[COURIER-2] DELIVERY_COMPLETE id=18 recipient=Pinar_Koc duration=1000ms
[COURIER-2] WAITING
[COURIER-8] DELIVERY_COMPLETE id=9 recipient=Hasan_Ozturk duration=1500ms
```

Figure 7: Scenario 5 high-concurrency output A: Beginning of the multiple-run high-concurrency scenario.


```

[COURIER-8] DELIVERY_COMPLETE id=9 recipient=Hasan_Ozturk duration=1500ms
[COURIER-8] WAITING
[COURIER-3] DELIVERY_COMPLETE id=12 recipient=Selin_Kaya duration=1500ms
[COURIER-3] WAITING
[COURIER-5] DELIVERY_COMPLETE id=15 recipient=Kemal_Sahin duration=1500ms
[COURIER-5] SHIFT_OVER
[COURIER-6] SHIFT_OVER
[COURIER-1] SHIFT_OVER
[COURIER-4] SHIFT_OVER
[COURIER-9] SHIFT_OVER
[COURIER-2] SHIFT_OVER
[COURIER-8] SHIFT_OVER
[COURIER-3] SHIFT_OVER
[COURIER-10] SHIFT_OVER
[COURIER-7] SHIFT_OVER
[CARGOGTU] SHIFT_END completed=20 cancelled=0 total_time=19500ms
[CARGOGTU] SHUTDOWN_COMPLETE
[CARGOGTU] SHIFT_START couriers=10 orders=20
[CARGOGTU] ORDER_QUEUED id=1 recipient=Ahmet_Yilmaz priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=4 recipient=Fatma_Sahin priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=7 recipient=Mustafa_Koc priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=10 recipient=Merve_Yildiz priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=13 recipient=Burak_Arslan priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=16 recipient=Rana_Yilmaz priority=EXPRESS duration=2
[CARGOGTU] ORDER_QUEUED id=19 recipient=Oguz_Yildiz priority=EXPRESS duration=3
[CARGOGTU] ORDER_QUEUED id=3 recipient=Mehmet_Demir priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=5 recipient=Ali_Celik priority=STANDARD duration=3
[CARGOGTU] ORDER_QUEUED id=8 recipient=Elif_Demir priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=11 recipient=Emre_Celik priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=14 recipient=Dilan_Ozkan priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=17 recipient=Tarik_Demir priority=STANDARD duration=1
[CARGOGTU] ORDER_QUEUED id=20 recipient=Ceren_Arslan priority=STANDARD duration=1
[CARGOGTU] ORDER_QUEUED id=2 recipient=Ayşe_Kaya priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=6 recipient=Zeynep_Arslan priority=ECONOMY duration=2
[CARGOGTU] ORDER_QUEUED id=9 recipient=Hasan_Ozturk priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=12 recipient=Selin_Kaya priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=15 recipient=Kemal_Sahin priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=18 recipient=Pinar_Koc priority=ECONOMY duration=2
[COURIER-2] DELIVERY_START id=1 recipient=Ahmet_Yilmaz priority=EXPRESS
[COURIER-1] DELIVERY_START id=4 recipient=Fatma_Sahin priority=EXPRESS
[COURIER-6] DELIVERY_START id=7 recipient=Mustafa_Koc priority=EXPRESS
[COURIER-4] DELIVERY_START id=10 recipient=Merve_Yildiz priority=EXPRESS
[COURIER-7] DELIVERY_START id=13 recipient=Burak_Arslan priority=EXPRESS
[COURIER-8] DELIVERY_START id=16 recipient=Rana_Yilmaz priority=EXPRESS
[COURIER-9] DELIVERY_START id=19 recipient=Oguz_Yildiz priority=EXPRESS
[COURIER-10] DELIVERY_START id=3 recipient=Mehmet_Demir priority=STANDARD
[COURIER-3] DELIVERY_START id=5 recipient=Ali_Celik priority=STANDARD
[COURIER-5] DELIVERY_START id=8 recipient=Elif_Demir priority=STANDARD
[COURIER-2] DELIVERY_COMPLETE id=1 recipient=Ahmet_Yilmaz duration=500ms
[COURIER-2] DELIVERY_START id=11 recipient=Emre_Celik priority=STANDARD
[COURIER-4] DELIVERY_COMPLETE id=10 recipient=Merve_Yildiz duration=500ms
[COURIER-4] DELIVERY_START id=14 recipient=Dilan_Ozkan priority=STANDARD
[COURIER-6] DELIVERY_COMPLETE id=7 recipient=Mustafa_Koc duration=500ms
[COURIER-6] DELIVERY_START id=17 recipient=Tarik_Demir priority=STANDARD
[COURIER-7] DELIVERY_COMPLETE id=13 recipient=Burak_Arslan duration=500ms
[COURIER-7] DELIVERY_START id=20 recipient=Ceren_Arslan priority=STANDARD
[COURIER-1] DELIVERY_COMPLETE id=4 recipient=Fatma_Sahin duration=500ms
[COURIER-1] DELIVERY_START id=2 recipient=Ayşe_Kaya priority=ECONOMY
[COURIER-8] DELIVERY_COMPLETE id=16 recipient=Rana_Yilmaz duration=1000ms
[COURIER-8] DELIVERY_START id=6 recipient=Zeynep_Arslan priority=ECONOMY
[COURIER-10] DELIVERY_COMPLETE id=3 recipient=Mehmet_Demir duration=1000ms
[COURIER-10] DELIVERY_START id=9 recipient=Hasan_Ozturk priority=ECONOMY
[COURIER-5] DELIVERY_COMPLETE id=8 recipient=Elif_Demir duration=1000ms
[COURIER-5] DELIVERY_START id=12 recipient=Selin_Kaya priority=ECONOMY
[COURIER-7] DELIVERY_COMPLETE id=20 recipient=Ceren_Arslan duration=500ms
[COURIER-7] DELIVERY_START id=15 recipient=Kemal_Sahin priority=ECONOMY
[COURIER-6] DELIVERY_COMPLETE id=17 recipient=Tarik_Demir duration=500ms
[COURIER-6] DELIVERY_START id=18 recipient=Pinar_Koc priority=ECONOMY
[COURIER-9] DELIVERY_COMPLETE id=19 recipient=Oguz_Yildiz duration=1500ms

```

Figure 8: Scenario 5 high-concurrency output B: Continuation of the high-concurrency output.

```

[COURIER-9] WAITING
[COURIER-4] DELIVERY_COMPLETE id=14 recipient=Dilan_Ozkan duration=1000ms
[COURIER-4] WAITING
[COURIER-2] DELIVERY_COMPLETE id=11 recipient=Emre_Celik duration=1000ms
[COURIER-2] WAITING
[COURIER-3] DELIVERY_COMPLETE id=5 recipient=Ali_Celik duration=1500ms
[COURIER-3] WAITING
[COURIER-1] DELIVERY_COMPLETE id=2 recipient=Ayse_Kaya duration=1500ms
[COURIER-1] WAITING
[COURIER-8] DELIVERY_COMPLETE id=6 recipient=Zeynep_Arslan duration=1000ms
[COURIER-8] WAITING
[COURIER-6] DELIVERY_COMPLETE id=18 recipient=Pinar_Koc duration=1000ms
[COURIER-6] WAITING
[COURIER-5] DELIVERY_COMPLETE id=12 recipient=Selin_Kaya duration=1500ms
[COURIER-5] WAITING
[COURIER-7] DELIVERY_COMPLETE id=15 recipient=Kemal_Sahin duration=1500ms
[COURIER-7] WAITING
[COURIER-10] DELIVERY_COMPLETE id=9 recipient=Hasan_Ozturk duration=1500ms
[COURIER-10] SHIFT_OVER
[COURIER-9] SHIFT_OVER
[COURIER-4] SHIFT_OVER
[COURIER-3] SHIFT_OVER
[COURIER-1] SHIFT_OVER
[COURIER-6] SHIFT_OVER
[COURIER-8] SHIFT_OVER
[COURIER-2] SHIFT_OVER
[COURIER-5] SHIFT_OVER
[COURIER-7] SHIFT_OVER
[CARGOGTU] SHIFT_END completed=20 cancelled=0 total_time=19500ms
[CARGOGTU] SHUTDOWN_COMPLETE
[CARGOGTU] SHIFT_START couriers=10 orders=20
[CARGOGTU] ORDER_QUEUED id=1 recipient=Ahmet_Yilmaz priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=4 recipient=Fatma_Sahin priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=7 recipient=Mustafa_Koc priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=10 recipient=Merve_Yildiz priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=13 recipient=Burak_Arslan priority=EXPRESS duration=1
[CARGOGTU] ORDER_QUEUED id=16 recipient=Rana_Yilmaz priority=EXPRESS duration=2
[CARGOGTU] ORDER_QUEUED id=19 recipient=Oguz_Yildiz priority=EXPRESS duration=3
[CARGOGTU] ORDER_QUEUED id=3 recipient=Mehmet_Demir priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=5 recipient=Ali_Celik priority=STANDARD duration=3
[CARGOGTU] ORDER_QUEUED id=8 recipient=Elif_Demir priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=11 recipient=Emre_Celik priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=14 recipient=Dilan_Ozkan priority=STANDARD duration=2
[CARGOGTU] ORDER_QUEUED id=17 recipient=Tarik_Demir priority=STANDARD duration=1
[CARGOGTU] ORDER_QUEUED id=20 recipient=Ceren_Arslan priority=STANDARD duration=1
[CARGOGTU] ORDER_QUEUED id=2 recipient=Ayse_Kaya priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=6 recipient=Zeynep_Arslan priority=ECONOMY duration=2
[CARGOGTU] ORDER_QUEUED id=9 recipient=Hasan_Ozturk priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=12 recipient=Selin_Kaya priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=15 recipient=Kemal_Sahin priority=ECONOMY duration=3
[CARGOGTU] ORDER_QUEUED id=18 recipient=Pinar_Koc priority=ECONOMY duration=2
[COURIER-1] DELIVERY_START id=1 recipient=Ahmet_Yilmaz priority=EXPRESS
[COURIER-2] DELIVERY_START id=4 recipient=Fatma_Sahin priority=EXPRESS
[COURIER-4] DELIVERY_START id=7 recipient=Mustafa_Koc priority=EXPRESS
[COURIER-5] DELIVERY_START id=10 recipient=Merve_Yildiz priority=EXPRESS
[COURIER-3] DELIVERY_START id=13 recipient=Burak_Arslan priority=EXPRESS
[COURIER-6] DELIVERY_START id=16 recipient=Rana_Yilmaz priority=EXPRESS
[COURIER-8] DELIVERY_START id=19 recipient=Oguz_Yildiz priority=EXPRESS
[COURIER-7] DELIVERY_START id=3 recipient=Mehmet_Demir priority=STANDARD
[COURIER-10] DELIVERY_START id=5 recipient=Ali_Celik priority=STANDARD
[COURIER-9] DELIVERY_START id=8 recipient=Elif_Demir priority=STANDARD
[COURIER-5] DELIVERY_COMPLETE id=10 recipient=Merve_Yildiz duration=500ms
[COURIER-5] DELIVERY_START id=11 recipient=Emre_Celik priority=STANDARD
[COURIER-1] DELIVERY_COMPLETE id=1 recipient=Ahmet_Yilmaz duration=500ms
[COURIER-4] DELIVERY_COMPLETE id=7 recipient=Mustafa_Koc duration=500ms
[COURIER-4] DELIVERY_START id=14 recipient=Dilan_Ozkan priority=STANDARD
[COURIER-3] DELIVERY_COMPLETE id=13 recipient=Burak_Arslan duration=500ms
[COURIER-2] DELIVERY_COMPLETE id=4 recipient=Fatma_Sahin duration=500ms
[COURIER-1] DELIVERY_START id=17 recipient=Tarik_Demir priority=STANDARD

```

Figure 9: Scenario 5 high-concurrency output C: Continuation of the high-concurrency output.

```

[COURIER-4] DELIVERY_START id=14 recipient=Dilan_Ozkan priority=STANDARD
[COURIER-3] DELIVERY_COMPLETE id=13 recipient=Burak_Arslan duration=500ms
[COURIER-2] DELIVERY_COMPLETE id=4 recipient=Fatma_Sahin duration=500ms
[COURIER-1] DELIVERY_START id=17 recipient=Tarik_Demir priority=STANDARD
[COURIER-3] DELIVERY_START id=20 recipient=Ceren_Arslan priority=STANDARD
[COURIER-2] DELIVERY_START id=2 recipient=Ayse_Kaya priority=ECONOMY
[COURIER-7] DELIVERY_COMPLETE id=3 recipient=Mehmet_Demir duration=1000ms
[COURIER-7] DELIVERY_START id=6 recipient=Zeynep_Arslan priority=ECONOMY
[COURIER-9] DELIVERY_COMPLETE id=8 recipient=Elif_Demir duration=1000ms
[COURIER-9] DELIVERY_START id=9 recipient=Hasan_Ozturk priority=ECONOMY
[COURIER-6] DELIVERY_COMPLETE id=16 recipient=Rana_Yilmaz duration=1000ms
[COURIER-6] DELIVERY_START id=12 recipient=Selin_Kaya priority=ECONOMY
[COURIER-1] DELIVERY_COMPLETE id=17 recipient=Tarik_Demir duration=500ms
[COURIER-1] DELIVERY_START id=15 recipient=Kemal_Sahin priority=ECONOMY
[COURIER-3] DELIVERY_COMPLETE id=20 recipient=Ceren_Arslan duration=500ms
[COURIER-3] DELIVERY_START id=18 recipient=Pinar_Koc priority=ECONOMY
[COURIER-8] DELIVERY_COMPLETE id=19 recipient=Oguz_Yildiz duration=1500ms
[COURIER-8] WAITING
[COURIER-10] DELIVERY_COMPLETE id=5 recipient=Ali_Celik duration=1500ms
[COURIER-10] WAITING
[COURIER-5] DELIVERY_COMPLETE id=11 recipient=Emre_Celik duration=1000ms
[COURIER-5] WAITING
[COURIER-4] DELIVERY_COMPLETE id=14 recipient=Dilan_Ozkan duration=1000ms
[COURIER-4] WAITING
[COURIER-7] DELIVERY_COMPLETE id=6 recipient=Zeynep_Arslan duration=1000ms
[COURIER-7] WAITING
[COURIER-2] DELIVERY_COMPLETE id=2 recipient=Ayse_Kaya duration=1500ms
[COURIER-2] WAITING
[COURIER-3] DELIVERY_COMPLETE id=18 recipient=Pinar_Koc duration=1000ms
[COURIER-3] WAITING
[COURIER-9] DELIVERY_COMPLETE id=9 recipient=Hasan_Ozturk duration=1500ms
[COURIER-9] WAITING
[COURIER-6] DELIVERY_COMPLETE id=12 recipient=Selin_Kaya duration=1500ms
[COURIER-6] WAITING
[COURIER-1] DELIVERY_COMPLETE id=15 recipient=Kemal_Sahin duration=1500ms
[COURIER-1] SHIFT_OVER
[COURIER-8] SHIFT_OVER
[COURIER-10] SHIFT_OVER
[COURIER-5] SHIFT_OVER
[COURIER-4] SHIFT_OVER
[COURIER-7] SHIFT_OVER
[COURIER-2] SHIFT_OVER
[COURIER-3] SHIFT_OVER
[COURIER-9] SHIFT_OVER
[COURIER-6] SHIFT_OVER
[CARGOGTU] SHIFT_END completed=20 cancelled=0 total_time=19500ms
[CARGOGTU] SHUTDOWN_COMPLETE
test5_run1.txt:[CARGOGTU] SHIFT_END completed=20 cancelled=0 total_time=19500ms
test5_run2.txt:[CARGOGTU] SHIFT_END completed=20 cancelled=0 total_time=19500ms
test5_run3.txt:[CARGOGTU] SHIFT_END completed=20 cancelled=0 total_time=19500ms
SHIFT_SUMMARY
Total orders      : 20
Completed         : 20
Cancelled         : 0
Total time        : 19500ms
Avg per order     : 975ms

COURIER_STATS
Courier-1 completed=3 total_time=2500ms
Courier-2 completed=2 total_time=2000ms
Courier-3 completed=3 total_time=2000ms
Courier-4 completed=2 total_time=1500ms
Courier-5 completed=2 total_time=1500ms
Courier-6 completed=2 total_time=2500ms
Courier-7 completed=2 total_time=2000ms
Courier-8 completed=1 total_time=1500ms
Courier-9 completed=2 total_time=2500ms
Courier-10 completed=1 total_time=1500ms

```

Figure 10: Scenario 5 high-concurrency output D: Final summaries, duplicate check, and final statistics file.

8 Challenges and Solutions

8.1 Challenge 1 — Preserving Priority Under Concurrency

The most important correctness problem was ensuring that no courier could take a lower-priority order while a higher-priority order was available. To solve this, all orders are inserted into a sorted priority queue before courier threads are started. Courier threads pop only from the head while holding `queue_lock`. This makes the selection atomic and prevents two couriers from taking the same order.

8.2 Challenge 2 — Clean SIGINT Shutdown

Signal handling in a multi-threaded C program can easily become unsafe if the signal handler prints, locks mutexes, or frees memory. The solution was to keep the signal handler minimal and only set a `sig_atomic_t` flag. The main thread performs the real shutdown sequence: it cancels pending queue entries, broadcasts to waiting couriers, waits for active deliveries to complete, joins all threads, and writes the final statistics.

8.3 Challenge 3 — Preventing Interleaved Logs

Multiple courier threads write to `stdout` at the same time. Without synchronization, their outputs could interleave and break the required output format. A dedicated `stdout_lock` protects every log line, including courier logs and CargoGTU coordinator logs.

9 Conclusion

The implementation satisfies the main assignment requirements: it uses a fixed thread pool, enforces priority scheduling, blocks idle couriers with condition variables, updates required counters with C11 atomic operations, handles `SIGINT` cleanly, and writes the required final statistics file.